

DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& also CS/IT Branch.

Linear Algebra

Basics of Determinants Matrices

Lecture No. **03**

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Recap of previous lecture



Topic

Algebra of Matrices



Topics to be Covered



Topic

Types of Matrices



Shortcut: Consider $A_{m \times n}$ & $B_{n \times p}$ then to find $(AB)_{m \times p}$

Number of Multiplications Required = $m \cdot n \cdot p$ & Number of Additions Req = $m(n-1)p$

eg if $A_{2 \times 3}$ & $B_{3 \times 3}$ then to find AB
 Here, $m=2, n=3, p=3$
 No. of Multi Req = $2 \times 3 \times 3 = 18$ Times
 No. of Additions Req = $2(3-1)3 = 12$ Times

eg if $A_{3 \times 4}$ & $B_{4 \times 3}$ then to find AB
 Here $m=3, n=4, p=3$
 No. of Multi Req = $3 \times 4 \times 3 = 36$ Times
 No. of Additions Req = $3(4-1)3 = 27$ Times

& to find BA
 Here $m=4, n=3, p=4$
 No. of Multi Req = $4 \times 3 \times 4 = 48$ Multiplications.
 No. of Additions Req = $4(3-1)4 = 32$ Additions

MAJEDAR QUESTION → Consider $A_{2 \times 3}$, $B_{3 \times 4}$ & $C_{4 \times 2}$ then find the minimum number of Multiplications & Additions that will be Required to find Matrix product (ABC) ?

Sol: To find Mat product (ABC) we have two Cases

Case I: $A_{2 \times 3} B_{3 \times 4} = (AB)_{2 \times 4}$ → No. of Multi = $2 \times 3 \times 4 = 24$
 No. of Additions = $2(3-1)4 = 16$

$(m=2, n=3, p=4)$

Now $(AB)_{2 \times 4} C_{4 \times 2} = (ABC)_{2 \times 2}$ → No. of Multi = $2 \times 4 \times 2 = 16$
 No. of Additions = $2 \times (4-1) \times 2 = 12$

$m=2, n=4, p=2$ i.e. By using Case I → No. of Multi = $24 + 16 = 40$
 Additions = $16 + 12 = 28$

Case II: $B_{3 \times 4} \begin{matrix} \searrow \\ 4 \times 2 \end{matrix} = (BC)_{3 \times 2} \begin{matrix} \nearrow \\ \text{Multi} = 3 \times 4 \times 2 = \boxed{24} \\ \searrow \\ \text{Add} = 3(4-1)2 = \boxed{18} \end{matrix}$
 $(m=3, n=4, p=2)$

Now $A_{2 \times 3} \begin{matrix} \searrow \\ 3 \times 2 \end{matrix} = (AC)_{2 \times 2} \begin{matrix} \nearrow \\ \text{Multi} = 2 \times 3 \times 2 = \boxed{12} \\ \searrow \\ \text{Addition} = 2(3-1)2 = \boxed{8} \end{matrix}$
 $m=2, n=3, p=2$

$\therefore$ No. of Multi $= 24 + 12 = 36$

$\&$ Addition $= 18 + 8 = 26$

$\therefore$ to Find Mat Product $(ABC) \begin{matrix} \nearrow \\ \text{Min Multi} = 36 \\ \searrow \\ \text{Min Addition} = 26 \end{matrix} \left. \vphantom{\begin{matrix} \nearrow \\ \text{Min Multi} = 36 \\ \searrow \\ \text{Min Addition} = 26 \end{matrix}} \right\} \underline{\underline{Ans}}$

Q $A = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}_{3 \times 1}$ & $B = [1 \ -1 \ 0]_{1 \times 3}$ & $C = \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix}_{3 \times 1}$ then $(ABC) = ??$

Sol: Case I: $AB = \begin{bmatrix} - \\ - \\ - \end{bmatrix}_{3 \times 3} [\dots] = \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}_{3 \times 3}$

$(AB)C = \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}_{3 \times 3} \begin{bmatrix} - \\ - \\ - \end{bmatrix}_{3 \times 1} = \begin{bmatrix} - \\ - \\ - \end{bmatrix}_{3 \times 1}$ Ans

Case II: $BC = [1 \ -1 \ 0]_{1 \times 3} \begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix}_{3 \times 1} = [2]_{1 \times 1}$

$(A)BC = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}_{3 \times 1} [2]_{1 \times 1} = \begin{bmatrix} 4 \\ -6 \\ 2 \end{bmatrix}$ Ans

Case II is comparatively easy.

If A & B are two matrices of same order then which of the following is true.

(a) $\times (A + B)^2 = A^2 + 2AB + B^2$

(b) $\times (A - B)^2 = A^2 - 2AB + B^2$

(c) $\checkmark (A + B)^2 + (A - B)^2 = 2A^2 + 2B^2$

(d) $\times (A + B)(A - B) = A^2 - B^2$

$\textcircled{d} (A+B)(A-B) = A^2 - \underbrace{AB+BA}_{??} - B^2$

$\textcircled{c} (A+B)^2 + (A-B)^2$

$= 2A^2 + 2B^2$

$\textcircled{a} (A+B)^2 = (A+B)(A+B)$
 $= A^2 + \underbrace{BA+AB}_{??} + B^2 \rightarrow 1$

$\textcircled{b} (A-B)^2 = (A-B)(A-B)$
 $= A^2 - \underbrace{AB-BA}_{??} + B^2 \rightarrow 2$

$\because AB \neq BA$ in general
 $\therefore AB+BA \neq 2AB$

Q. If $\boxed{XY=Y}$ & $\boxed{YX=X}$ then $X^2+Y^2=?$

MSB

- (a) $X+Y$
- (b) $2I$
- (c) O
- (d) I

Case I: Let X & Y both are Non Sing.

ie X^{-1} & Y^{-1} exist. So

$$XY=Y \Rightarrow X=I$$

$$\& YX=X \Rightarrow Y=I$$

$$\begin{aligned} \text{So } X^2+Y^2 &= I^2+I^2 \\ &= I+I \\ &= 2I \end{aligned}$$

Case II: Let X & Y both are Sing
ie X^{-1} & Y^{-1} DNE.

$$\begin{aligned} \text{So } X^2+Y^2 &= XX+YY \\ &= X(YX)+Y(XY) \\ &= (XY)X+(YX)Y \\ &= (Y)X+(X)Y \\ &= X+Y \\ &= \text{(a)} \checkmark \end{aligned}$$

Complex Conjugate: $z = x + iy$ then $\bar{z} = x - iy$ for $z = 2 - 3i, \bar{z} = 2 + 3i$

Conjugate Mat if $A = \begin{bmatrix} 2 & 4-i & 2i \\ -2+i & 0 & -5 \end{bmatrix}_{2 \times 3}$
 then $A^T = \begin{bmatrix} 2 & -2+i \\ 4-i & 0 \\ 2i & -5 \end{bmatrix}_{3 \times 2}$ $\bar{A} = \begin{bmatrix} 2 & 4+i & -2i \\ -2-i & 0 & -5 \end{bmatrix}_{2 \times 3}$

$$z = 2 + 4i, \bar{z} = 2 - 4i$$

$$z = -4i, \bar{z} = 4i$$

$$z = 2i, \bar{z} = -2i$$

$$z = 4, \bar{z} = 4$$

$$z = -3, \bar{z} = -3$$

$$z = 0, \bar{z} = 0$$

Transjugate Mat (Transposed Conjugate of Mat)

$$A^0 = (\bar{A}^T) \text{ or } (\bar{A})^T \text{ So } A^0 = \begin{bmatrix} 2 & -2-i \\ 4+i & 0 \\ -2i & -5 \end{bmatrix}_{3 \times 2}$$

(*) the concept of Transpose Mat, Conjugate Mat & Transjugate Mat are also defined for Rectangular Matrices

⊛ The Concept of Trace, Det, Inverse are defined only for square Matrices
while the " of Transpose, Conjugate, Transjugate & Rank
 are also defined for Rectangular Matrices

Real Mat: If in a Mat all the elements are Real Numbers then A is called Real Matrin
 i.e. if A is a Real Mat then $\overline{A} = A$ or $A^{\theta} = A^T$ Learn.

Complem Mat: If in a Mat, at least one element is Complem Number then
 Mat is called Complem Mat i.e. $\overline{A} \neq A$

e.g. $A = \begin{bmatrix} 2 & -5 & 4 \\ 1 & 2 & 0 \end{bmatrix} = \text{Real Mat}$, $B = \begin{bmatrix} 2 & -5 & 0 \\ 1 & 2 & 4i \end{bmatrix} = \text{Complem Mat}$

Reversal Law $\rightarrow$ $(ABC)^T = C^T B^T A^T$, $(ABC)^0 = C^0 B^0 A^0$, $(ABC)^{-1} = C^{-1} B^{-1} A^{-1}$ 

& these Results can also be generalized.

Qe if A, B, C, D are Non singular Matrices of same order $n \times n$

$ABCD = I$ then evaluate B & $B^{-1} = ?$

Sol: (M-I) $B = \frac{I}{ACD} = I \bar{A} \bar{C} \bar{D} = \bar{A} \bar{C} \bar{D}$

Blunder

$$(BC) \bar{C} = \bar{A} \bar{D} \bar{C}$$

$$BI = \bar{A} \bar{D} \bar{C}$$

$$B = \bar{A} \bar{D} \bar{C} \quad \underline{\underline{Ans}}$$

(M-II)

$$ABCD = I$$

$$\bar{A} (ABCD) \bar{D} = \bar{A} \cdot I \bar{D}$$

$$I B C I = \bar{A} \bar{D}$$

$$BC = \bar{A} \bar{D}$$

(ii) $B^{-1} = (\bar{A} \bar{D} \bar{C})^{-1}$
 $= (\bar{C})^{-1} (\bar{D})^{-1} (\bar{A})^{-1}$ (By R-Law)
 $B^{-1} = C D A \quad \underline{\underline{Ans}}$

Doubt

No. System

$$1 \times n \times 1 = 2$$

$$n \times 1 = 2$$

$$n = 2$$

Mat. Algebra

$$IAB = C$$

$$ABI = C$$

$$AB = C$$

for eg $A = \begin{bmatrix} 2 & 3 & -1 \\ 4 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$

then

$$IA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & -1 \\ 4 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 3 & -1 \\ 4 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix} = A$$

$$AI = \begin{bmatrix} 2 & 3 & -1 \\ 4 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 & -1 \\ 4 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix} = A$$

Sp. Points: (Some confusions) →



$$\textcircled{1} \text{Tr}(A+B+C) = \text{Tr}(A) + \text{Tr}(B) + \text{Tr}(C)$$

$$\textcircled{2} (A+B+C)^{\theta} = A^{\theta} + B^{\theta} + C^{\theta}$$

$$\textcircled{3} (A+B+C)^T = A^T + B^T + C^T$$

$$\textcircled{4} |A+B+C| \neq |A| + |B| + |C|$$

$$\textcircled{5} (A+B+C)^{-1} \neq A^{-1} + B^{-1} + C^{-1}$$

$$\text{But } (A+B+C)^{-1} = \frac{\text{adj}(A+B+C)}{|A+B+C|}$$

$$\textcircled{6} |ABC| = |A| \cdot |B| \cdot |C|$$

REVERSAL LAW: →

$$(ABC)^T = C^T B^T A^T$$

$$(ABC)^{\theta} = C^{\theta} B^{\theta} A^{\theta}$$

$$(ABC)^{-1} = C^{-1} B^{-1} A^{-1}$$

⊗ we know that, $AB \neq BA$ in general
But $\text{Tr}(AB) = \text{Tr}(BA)$ always.
(if both exist)

Note: if $A_{n \times n}$ then evaluate $|A \cdot \text{adj} A|$, $|\text{adj} A|$, $|\text{cof} A|$, $|A^{-1}| = ?$

So: w.k. that $A^{-1} = \frac{\text{adj} A}{|A|}$

$$A A^{-1} = \frac{A \cdot \text{adj} A}{|A|}$$

$$I = \frac{A(\text{adj} A)}{|A|}$$

$$\text{or } \boxed{A \cdot \text{adj} A = |A| \cdot I_n}$$

$$\text{ie } A(\text{adj} A) = |A| \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix}_{n \times n}$$

$$\text{ie } A(\text{adj} A) = \begin{bmatrix} |A| & 0 & 0 & \dots & 0 \\ 0 & |A| & 0 & \dots & 0 \\ 0 & 0 & |A| & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & |A| \end{bmatrix}_{n \times n} = \text{Scalar Mat.}$$

So $\det(A \cdot \text{adj} A) = \text{Product of diag. elements}$

$$|A(\text{adj} A)| = \underbrace{(|A|)(|A|)(|A|) \dots (|A|)}_{n \text{ times}}$$

$$\boxed{|A \cdot \text{adj} A| = |A|^n} \quad \text{Ans of Part (i) (MAG)}$$

$$(ii) \because |A(\text{adj} A)| = |A|^n$$

$$|A| \cdot |\text{adj} A| = |A|^n$$

$$|\text{adj} A| = \frac{|A|^n}{|A|}$$

$$\boxed{|\text{adj} A| = |A|^{n-1}}$$

ie Ans of part (2)

$$(iii) \text{ w.k. that } |\text{Any Mat}| = |\text{it's Transpose Mat}|$$

$$\text{so } |(\text{Cof} A)| = |(\text{Cof} A)^T| = |(\text{adj} A)| = |A|^{n-1}$$

$$\text{so } \boxed{|\text{Cof} A| = |A|^{n-1}}$$

$$(iv) \because A^{-1} = \frac{\text{adj} A}{|A|}$$

$$\text{so } |A^{-1}| = \left| \left(\frac{\text{adj} A}{|A|} \right) \right| = \frac{1}{|A|^n} |\text{adj} A|$$

$$= \frac{1}{|A|^n} (|A|^{n-1}) = \left(\frac{1}{|A|} \right)$$

$$\text{ie } \det(A^{-1}) = \frac{1}{\det(A)}$$

Ans of part (iv)
(BMA4)

Ans of part (iii)

Generalised formula: if $A_{n \times n}$ then

$$\underbrace{|\text{adj adj adj} \dots \text{adj} A|}_{r \text{ times}} = |A|^{(n-1)^r} \quad (\text{Learn})$$

for eg $|\text{adj} A| = |A|^{(n-1)^1}$, $|\text{adj adj} A| = |A|^{(n-1)^2}$ & so on.....

if $|A|_{5 \times 5} = 2$ then $|\text{adj adj adj} A| = ? = |A|^{(n-1)^3}$
 $n=5$ $r=3$

- (a) 2^{12}
- (b) 2^{64}
- (c) 2^{32}
- (d) 16^2

Note: $(2^4)^3 = 2^4 \times 2^4 \times 2^4 = 2^{12} \checkmark$
 $(2^{4^3}) = 2^{4 \times 4 \times 4} = 2^{64} \checkmark$

$$= |A|^{(5-1)^3} = |A|^{4^3} = (2)^{4^3} = (2)^{64}$$

Some More Special Types of Matrices →



eg $\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$

(Symmetric)
($\because A^T = A$)

$\begin{bmatrix} 0 & h & -g \\ -h & 0 & f \\ g & -f & 0 \end{bmatrix}$

(Skew Symm)
($\because A^T = -A$)

$\begin{bmatrix} 2 & 4+i & -5i \\ 4-i & 0 & 7 \\ 5i & 7 & -3 \end{bmatrix}$

(Hermitian Mat)
($\because A^\theta = A$)

$\begin{bmatrix} 2i & 4+i & -5i \\ -4+i & 0 & 7 \\ -5i & -7 & -3i \end{bmatrix}$

(skew Hermitian)
($\because A^\theta = -A$)

- ⊗ For Any Mat $A_{n \times n}$, AA^T & $A^T A$ are always Symmetric, $n = \text{odd}$
 - ⊗ if $A_{n \times n}$ s.t A is skew symm then $|A| = \begin{cases} 0 & n = \text{odd} \\ \text{Perfect square}, & n = \text{even} \end{cases}$
- ie Det of skew symm Mat of odd order is always Zero

- But Det of skew symm Mat of even order is Perfect sq. or may be Zero also
- ⊗ Sum of all the elements of skew symm Mat is always Zero

eg: $A = \begin{bmatrix} 0 & -5 & 3 & 2 & -1 \\ 5 & 0 & 4 & 0 & -2 \\ -3 & -4 & 0 & 4 & -1 \\ -2 & 0 & -4 & 0 & 3 \\ 1 & 2 & 1 & -3 & 0 \end{bmatrix}_{5 \times 5}$

= Skew Sym Mat of odd order

So $|A| = 0$

eg: $A = \begin{bmatrix} 0 & 1 & 2 & 0 \\ -1 & 0 & 4 & 0 \\ -2 & -4 & 0 & 6 \\ 0 & 0 & -6 & 0 \end{bmatrix}_{4 \times 4}$

= skew sym Mat of even order then $|A| = ?$

- (a) 3 (b) 36 (c) 15 (d) 41

if options are (a) 3 (b) 36 (c) 16 (d) 0 then calculate using conventional approach

eg: $A = \begin{bmatrix} 0 & -1 & -2 & -3 \\ 1 & 0 & -1 & -2 \\ 2 & 1 & 0 & -1 \\ 3 & 2 & 1 & 0 \end{bmatrix}_{4 \times 4}$

then $|A| = ?$ (a) 16 (b) 49 (c) 0, (d) 5

Using Conventional Approach, $|A| = \dots = 0$

① Let $A_{m \times n}$ Then $\begin{cases} (AA^T)_{m \times m} \rightarrow \text{Symmetric Mat} \\ (A^T A)_{n \times n} \rightarrow \text{Symmetric Mat} \end{cases}$

eg $A = \begin{bmatrix} 2 & -1 & 4 \\ 1 & 0 & 2 \end{bmatrix}_{2 \times 3}$ Then $AA^T = \begin{bmatrix} 21 & 10 \\ 10 & 5 \end{bmatrix}_{2 \times 2} = \text{Symm.}$

Then $A^T = \begin{bmatrix} 2 & 1 \\ -1 & 0 \\ 4 & 2 \end{bmatrix}_{3 \times 2}$

$A^T A = \begin{bmatrix} 5 & -2 & 10 \\ -2 & 1 & -4 \\ 10 & -4 & 20 \end{bmatrix}_{3 \times 3} = \text{Symm.}$

Doubt

$A =$

$$\begin{bmatrix} 2i & - & - \\ - & 0 & - \\ - & - & -5i \end{bmatrix}$$

where

$$a_{ij} = -\overline{a_{ji}}$$

At $(i=j)$

$$a_{ii} = -\overline{a_{ii}}$$

$$a_{ii} + \overline{a_{ii}} = 0$$

$$\text{let } a_{ii} = 2 \Rightarrow 2 + 2 \neq 0 \quad \times$$

$$\text{let } a_{ii} = 0 \Rightarrow 0 + 0 = 0 \quad \checkmark$$

$$\text{let } a_{ii} = 2i \Rightarrow 2i + (2i) = 0 \quad \checkmark$$

let $a_{ii} = 4 + 3i$ then

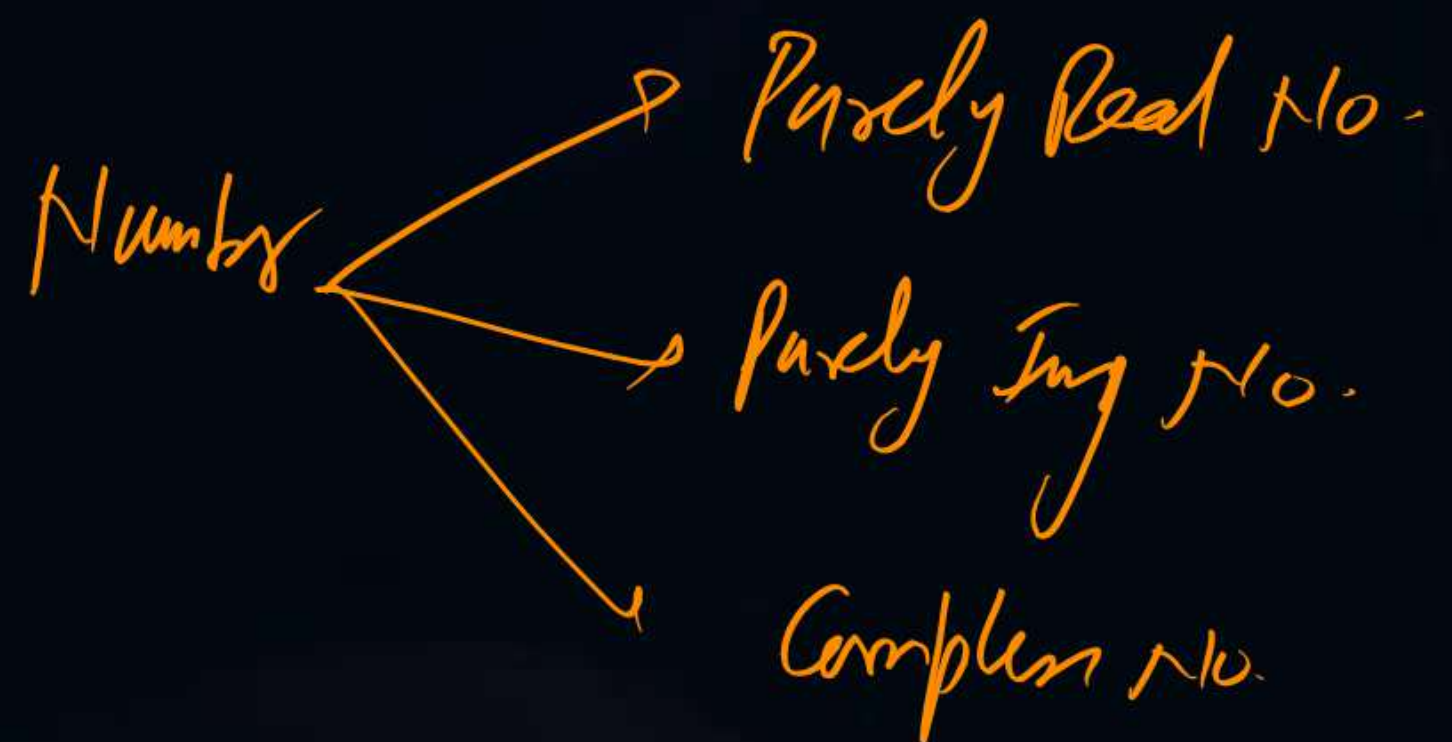
$$a_{ii} + \overline{a_{ii}} = 4 + 3i + 4 - 3i$$

$$= 8$$

$$\neq 0$$

Doubt

Number



$$z = 2 = 2 + 0i$$

$$z = 2i = 0 + 2i$$

$$z = 2 + 3i$$

Qe If $A = [a_{ij}]_{n \times n}$ s.t. $a_{ij} = i^2 - j^2 + i \delta_{ij}$ for $n \geq 3$, $n = \text{odd}$, $A^{-1} = ?$

Ans: possible values of n are 3, 5, 7, 9, 11, ...

Let $n=3$, $A = [a_{ij}]_{3 \times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 0 & -3 & -8 \\ 3 & 0 & -5 \\ 8 & 5 & 0 \end{bmatrix}$ = skew Symm of odd order 3×3

So $|A| = 0$ i.e. A is singular so $A^{-1} = \text{DNE}$

Qe if $A = [a_{ij}]_{n \times n}$ s.t. A is skew Symm of order n then $\sum a_{ij} = ?$

- (a) 0
(b) n
(c) $n-1$
(d) $\frac{n^2-n}{2}$
- $\therefore$ for skew Symm Mat, Diag elements are all zero $\forall i \neq j$
& corresponding elements are of opposite sign Hence $\sum_{\forall i \neq j} a_{ij} = 0$
- $$\sum a_{ij} = \sum a_{ii} + \sum a_{ij} + \sum a_{ji}$$
- $$= 0 + \sum a_{ij} - \sum a_{ij} = 0$$

Q If A & B are Hermitian Mat then $(AB-BA)$ will be?

- (a) Symm.
- (b) Skew Symm
- (c) Hermitian
- (d) ☒ Skew Hermitian

Given $A^\theta = A$ $\xrightarrow{(1)}$ $B^\theta = B$ $\xrightarrow{(2)}$

$$\begin{aligned}
 \text{Now, } (AB-BA)^\theta &= (AB)^\theta - (BA)^\theta \\
 &= B^\theta A^\theta - A^\theta B^\theta \quad (\text{By R. Law}) \\
 &= BA - AB \\
 &= -(AB-BA)
 \end{aligned}$$

So $(AB-BA)$ will be skew Hermitian.

A, B be any $n \times n$ matrices. Then choose incorrect statement.

- (a) AA^T is symmetric (T)
- (b) $A + A^T$ is symmetric (T)
- (c) $A^T A$ is symmetric (T)
- (d) $A - A^T$ symmetric (F)

(d) let $S = A - A^T$
 then $S^T = (A - A^T)^T = A^T - (A^T)^T = A^T - A$
 $= -(A - A^T) = -S$
 i.e. $S^T = -S \Rightarrow S$ is skew symmetric
 or $(A - A^T)$ is skew symmetric.

(a) let $P = AA^T$ then
 $P^T = (AA^T)^T = (A^T)^T A^T$
 $= AA^T$
 i.e. $P^T = P$ so P is symmetric.

(c) let $Q = A^T A$
 $Q^T = (A^T A)^T = A^T (A^T)^T$
 $= A^T A = Q$
 $\therefore Q^T = Q$ so $A^T A$ is symmetric.

(b) let $R = A + A^T$ then $R^T = (A + A^T)^T = A^T + (A^T)^T = A^T + A = R$ so R is symmetric.
 i.e. $A + A^T$ is symmetric.

(5) Every sq. Mat can be expressed as the sum of Symm & Skew Symm Mat. as follows; Consider sq Mat $A_{n \times n}$ Then by common sense we can write.

$$A = \left(\frac{A+A^T}{2} \right) + \left(\frac{A-A^T}{2} \right)$$

$A = P + Q$ where $P = \left(\frac{A+A^T}{2} \right)$ & $P^T = P$ i.e. P is Symm.
 $Q = \left(\frac{A-A^T}{2} \right)$ & $Q^T = -Q$ i.e. Q is skew Symm.

$$\text{sq. Mat} = \text{Symm} + \text{Skew Symm}$$

$$A = \left(\frac{A+A^T}{2} \right) + \left(\frac{A-A^T}{2} \right)$$

(5) Every sq. Mat can be expressed as the sum of Hermitian & Skew Herm Mat. as follows; Consider sq Mat $A_{n \times n}$ Then by common sense we can write.

Pr:

$$A = \left(\frac{A + A^{\theta}}{2} \right) + \left(\frac{A - A^{\theta}}{2} \right)$$

$R = \left(\frac{A + A^{\theta}}{2} \right)$ where $R^{\theta} = R$ i.e. R is Hermitian

$$A = R + S \quad \text{i.e.}$$

$S = \left(\frac{A - A^{\theta}}{2} \right)$ where $S^{\theta} = -S$ i.e. S is Skew Herm.

i.e. $\boxed{\text{Sq. Mat} = \text{Herm Mat} + \text{Skew Herm. Mat}}$

where $R^{\theta} = \left(\frac{A + A^{\theta}}{2} \right)^{\theta} = \frac{A^{\theta} + (A^{\theta})^{\theta}}{2} = \frac{A^{\theta} + A}{2} = R$ i.e. R is Herm. Mat

& $S^{\theta} = \left(\frac{A - A^{\theta}}{2} \right)^{\theta} = \frac{A^{\theta} - (A^{\theta})^{\theta}}{2} = \frac{A^{\theta} - A}{2} = -\left(\frac{A - A^{\theta}}{2} \right) = -S$ i.e. S is Skew, Hermitian

Q.2 $A = \begin{bmatrix} 0 & 2-3i \\ -2-3i & 0 \end{bmatrix}$ then A is ? $(n \times n) \therefore A^0 = (\overline{A^T})$

(a) Symm

(b) skew symm

(c) Hermitian

(d) Skew Hermitian

$$a_{12} = 2-3i$$

$$\overline{a_{12}} = 2+3i$$

$$\text{So } -\overline{a_{12}} = -2-3i = a_{21}$$

$$= -A$$

for A is skew H.

$$A^T = \begin{bmatrix} 0 & -2-3i \\ 2-3i & 0 \end{bmatrix}, \quad A^0 = (\overline{A^T}) = \begin{bmatrix} 0 & -2+3i \\ 2+3i & 0 \end{bmatrix} = -\begin{bmatrix} 0 & 2-3i \\ -2-3i & 0 \end{bmatrix} = -A$$



(Dr Puneet Sirpw)



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Thank
YOU